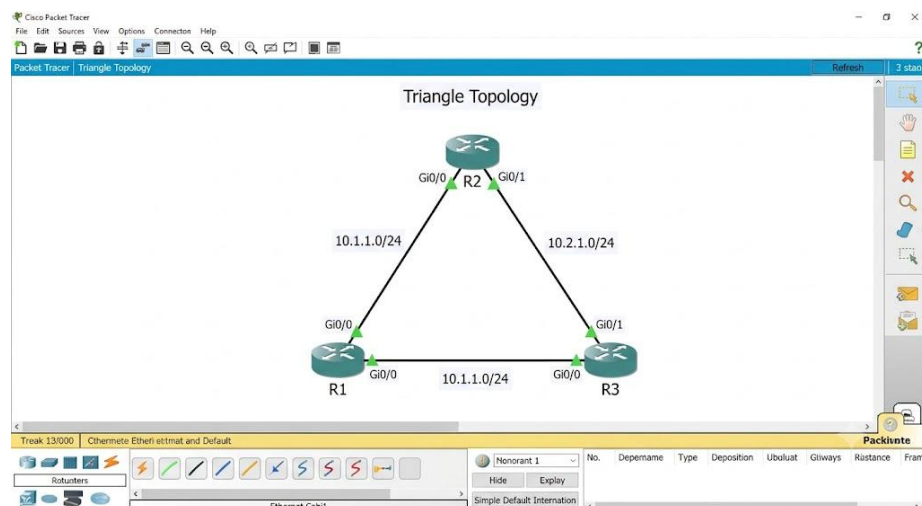


Day 4: Introduction to OSPF Theory

Task 1 :- Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

ANS :-



Create the Triangle Topology :-

1. Place Routers : Drag 3 Routers (like 4331) onto the canvas, forming a triangle (R1, R2, R3).

2. Cable Them : Connect R1 → R2, R2 → R3, and R3 → R1 using Copper Cross-Over (or Serial) cables.
3. Activate Interfaces : Access the CLI of each router and turn on all connecting interfaces :

```
en
conf t
int g0/0/0 (Adjust for your connections, e.g., G0/0/1, S0/1/0)
no shut
```

Task 2 :- Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

ANS :-

Set IP Addresses:

Router 1:

```
en
conf t
int g0/0/0
ip address 10.1.1.1 255.255.255.0
no shut
```

Router 2:

```
en
conf t
int g0/0/0
ip address 10.1.1.2 255.255.255.0
no shut
```

Verify Connectivity (Ping):

- Open Router 1 CLI and type:

```
ping 10.1.1.2
```

Success Check : You should see Success rate is 80 or 100 percent with exclamation points (!!!!!).

Task 3 :- Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

ANS :-

Enable OSPF on Each Router

Open each router's CLI and run these commands (matching your assigned subnets):

On R1:

```
en
conf t
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 10.3.3.0 0.0.0.255 area 0
```

On R2 :

```
en
conf t
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
```

On R3 :

```
en
conf t
router ospf 1
network 10.2.2.0 0.0.0.255 area 0
network 10.3.3.0 0.0.0.255 area 0
```

Check Success

Type show ip route on any router. Routes learned via OSPF will show up with an **O!**

Task 4 :- Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router.

ANS :-

1.Type command on each router :

```
show ip ospf neighbor
```

2.Check Neighbor ID & State: Look at the State column—it should say FULL (e.g., FULL/DR or FULL/BDR).

3.What FULL means: Routers are fully synced and talking to each other.

4.The 3 main steps it goes through:

- 2-WAY: Routers discover each other.
- LOADING: Exchanging route info.
- FULL: Neighbor setup complete!

Task 5 :- Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table.

ANS :-

1.Change Cost on an Interface (e.g., R1's G0/0/0) :

```
en
conf t
interface g0/0/0
ip ospf cost 50
```

2.Verify the Effect :

```
show ip route
```

What Changes :-

- Higher Total Metric : The cost for routes going through that interface increases .
- Path Shift : If a cheaper alternate path exists , OSPF automatically switches to that faster/lower-cost route.